

Strategies of Case-marking on Classical Greek complement clauses

A plea for null expletives¹

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Abstract — Participation in the derivation of argumental DPs is tightly related to their ability to be Case-marked. In contrast, Finite Complement Clauses seem to be reluctant to Case-marking. Notwithstanding, they can function as arguments. This article addresses this paradox from the perspective of Classical Greek. It shows that its FCCs are nominals endowed with ϕ -features, which do value the features of functional heads like v° or T° , as expected from *bona fide* arguments. However, they are Case-less because they do not project a CaseP, so they cannot be directly tied to v° or T° , which is also reflected in their more restricted placement, compared to run-of-the-mill DPs. Which is why they make use of proxy strategies to become active in the derivation. This indirect participation in the derivation necessitates positing null expletives for uniformity in an array of cases. Strong independent motivations for null expletives are adduced.

1. Introduction

1.1 Case

This article is about Finite Complement Clauses (FCCs) and their way of getting Case, although they look deprived of it, although they exhibit the properties of nominal arguments.

Case is a morphological or abstract marking on an NP (or a DP), which depends on syntactic (subjecthood, objecthood) and interpretive (semantic role) properties. This marking can be conceived of as the valuation of the Case-feature that any DP carries (Chomsky 2001). Alternatively, DPs can be viewed as born with a large structure representing the whole range of case possibilities and the necessary subpart of it is extracted (Peeling theory, Caha 2009).² In the present article, I am not so much interested in the morphology of case and how it arises or in the difference between structural/syntactic and inherent/semantic (and lexical) Case³ as in the proximity between grammatical Case and agreement in the process of deriving a clause.⁴ In particular, I will focus on the relationship between Case-assigning head (typically verbs, tenses, prepositions) and their goals.

1.2 Case and subordinate clauses

This ability of DPs to be Case-marked goes hand in hand with their possibility/necessity of playing a semantic and syntactic role with respect to Case-assigners. The phrases that non-nominal categories, like adverbs, verbs, tenses or complementizers, project are normally not assigned Case because they do not play such semantic roles. Nonetheless, there exist

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² There are many more theories of Case, most of which are reviewed in Mal'čukov and Spencer (2009). Here, I concentrate on generative theories. In the conclusion of Section 3.2.1, I will say a few words about the dependent Case theory (Marantz 1991) and why I do not adopt it.

³ See Woolford (2006), a.o.

⁴ I abide by the tradition of writing the word *Case* with a capital when referring to grammatical Case (which can be abstract) and with a minuscule when referring to morphological case.

exceptions to this stance, especially among subordinate clauses. For example, adverbial clauses can be analyzed as Case-marked CPs (cf. most articles in this volume). Likewise, relative clauses also appear Case-marked sometimes, especially free relatives. Consider for example Plato's excerpt in (1).⁵

- (1) Classical Greek, Pl. *Grg.* 501a2

Éskeptai tèn aitían [RC **hôn** **práttei**].
 has.investigated the cause what:GEN.PL it.does
 '(The medicine) has investigated the cause of its practice (lit. of what it does).'

The relative clause *hôn práttei* 'what it does' complements the noun *aitían* 'cause'. In the relative clause, the verb *práttei* 'it.does' assigns accusative, so the relative pronoun that is its object should be in the accusative (*há* 'what.acc'). Instead, it is in the genitive because the whole relative is assigned genitive. Indeed, the clause complements a noun and for this reason is assigned genitive, a Case which percolates up to and is imposed on the relative pronoun. This marking should come as no surprise, since free relatives can or must distribute like DPs, depending on the theory (Bresnan and Grimshaw 1978; Jacobson 1995; de Vries 2002; Caponigro 2003; Donati 2006).

We are left with the third type of subordinate clause, namely argument clauses, which (2)-(5) illustrate.

- (2) Russian (Knyazev 2016: 1)

Učenyje govorjat, čto na ètoj territorii ran'se žili ljudi.
 scientists:NOM say that on this territory:LOC earlier lived people:NOM
 'Scientists say that earlier people used to live on this territory.'

- (3) Classical Greek, X. *HG* 1.6.32

Hérmōn (...) eípe pròs autòn **hóti eíē kalôs échon apopleûsai**.
 H.NOM said to him that was well having to.sail.away
 'Hermon said to him it was well to sail away.'

⁵ The Classical Greek morphology is extremely rich but not agglutinative. To keep the examples readable, I refrain from segmenting the morphemes and glossing all the features. Instead, I use approximate English translations and gloss the necessary parts only (which may vary from one example to another) according to the Leipzig Glossing Rules of the Max Planck Institute of Evolutionary Anthropology and the Department of Linguistics of the University of Leipzig, to which I added AOR for aorist. The abbreviations for Greek authors and works are those of Liddell and Scott (1996).

(4) Russian (Knyazev 2016: 2)

Èti naxodki govorjat o tom,

these findings:NOM say about it:LOC

čto na ètoj territorii ran'she žili ljudi.

that on this territory:LOC earlier lived people:NOM

‘These findings indicate that earlier people used to live on this territory.’

(5) Korean (Bogal-Allbritten and Moulton 2018: 220, glosses as in original)

Na-nun [ney-ka swukecey-lul ta ha-yess-ta-nun kes-ul] mit-e.

I-TOP you-NOM homework-ACC all do-PST-DEC-ADN kes-ACC believe-DEC

‘I believe (the claim) that you finished your homework.’

As arguments, they too are likely to pattern with DPs, in that they are endowed with a θ -role attributed by a verb, and thus are expected to receive the reflex of this θ -role attribution, namely Case. However, crosslinguistically, they do not as shown for example by Russian (2) and Classical Greek (3). When they do, they are considered nominalized, so that it is a determiner, as *tom* in (4), or a nominal head, as *kes* in (5) receives Case (see the glosses) and not the CP itself.

1.3 Caseless argument clauses and syntactic theory

This inability to be Case-marked becomes even more mysterious from a theoretical perspective. In generative grammar, the role of Case is peripheral: A DP can be recruited to fulfill some requirements such as being endowed with a θ -role or value ϕ -features on T° or v° . We have no indication that θ -role assignment does not take place. In fact, if we follow Rouveret's (2023) analysis, their being featurally (and tense) complete makes FCCs inherently visible for this θ -role assignment, say by a V° within vP (see Section 5.1).

The problem comes from the relation with T° and v° , and ϕ -feature valuation thereof. For T° and v° to be able to probe for it and enter in a relationship (agree) with it, a DP must be active and visible. Its carrying an unvalued Case-feature is the condition for this activation condition (Chomsky 1981; 1995; 2000, a.o.), since it lacks something and is thus kept “awake.” As arguments, FCCs are expected to behave similarly (be active) and thus end up Case-marked, contrary to fact.

Thus, we are facing a paradox. The lack of deficiency which makes FCCs visible and permits θ -role assignment prevents them from being active for the operation of ϕ -feature valuation. At the heart of the present article is the question as to how the derivation can converge despite these difficulties.

1.4 Case and “extraposition”

The prevention against Case-marking may be directly reflected in their tendency to look extraposed, as influentially observed by Stowell (1981). A recent thread of thought goes a step further and argues that complement clauses are not arguments at all, but adjunct-like

(Moulton 2009; Bondarenko 2022, based on an original proposal by Kratzer 2006), which explains why they are peripheral and Case-less.

However, it is beyond its scope to discuss the on-going argument-vs.-adjunct debate. Instead, it will assume that complement clauses are arguments, because they are obligatory and are in a selectional relationship with the verbs as shown by the reverse pattern that (6) and (7) exhibit, famously already observed in Grimshaw (1979). These two features are unexpected under the clauses-as-adjuncts view.⁶

(6) Peter and Mendy asked *(if the weather was nice)/(that the weather was nice).

(7) Peter and Mendy regretted (*if the weather was nice)/*(that the weather was nice).

1.5 Claims

The present article aims to reassess the above-reported theories and facts (θ -assignment, ϕ -feature valuation, “extraposition”) with the idea that they must be explained together, as several faces of the same object. It will be claimed that lack of Case is central in accounting for the behavior of FCCs but not for the reasons that were raised so far in the literature (like Stowell's 1981 Case Resistance Principle). The investigation will be based on a thorough examination of Classical Greek facts,⁷ as well as mentions of other languages. Three hypotheses will be explored:

1. Are FCCs covertly Case-marked?
2. Are FCCs totally Case-less?
3. Are FCCs indirectly Case-marked? (e.g., as part of a chain)

The following claims will be made, in support to hypothesis 3:

- FCCs are DPs;
- FCCs are Case-less (no CaseP is projected);
- FCCs can play a role only remotely, via proxies or post-syntactically;
- The proxy can be null expletive.

The empirical contribution rests in the evidence for ϕ -feature carrying by FCCs and for their ϕ -agreement with v° and T° . The theoretical contributions are the severing of DP-hood from Case-marking and the support for the existence of subject and object null expletives.

1.6 Outline

The article is organized as follows. In Section 2, I show that Classical Greek FCCs are some kind of DPs, which carry ϕ -features. Section 3 discusses the position of FCCs within the clause and argues that they are *in situ*, i.e., where they were first merged, in a θ -assigning position. Section

⁶ The debate is also grounded in the semantics of complement clauses, some scholars arguing that they are entities and denote Vendlerian objects (events, facts, propositions, cf. Vendler 1967; Zucchi 1993; Faure 2021, submitted), while other thinks that they are predicates (Kratzer 2006; Moulton 2015; Elliott 2020; Bondarenko 2022; Moltmann 2024). Finally, some researchers consider that both denotations are needed, depending on the situation (Djärv 2019; Özyıldız 2020).

⁷ Classical Greek is Attic Greek spoken in the Fifth and Fourth C. BCE. The data come from a corpus study based on Aeschylus' (b. c. 525 BCE) *Persae*, *Supplices*, Sophocles' (b. c. 497) *Philoctetes*, Thucydides' (b. c. 460) *The Peloponnesian War* (books 1-4), Lysias' (b. c. 445) *Speeches* 1-15, Xenophon's (b. c. 430) *Hellenica* (books 1-3). I concentrate on one type of FCCs, namely the one introduced by the complementizer *hóti* (466 examples in the corpus), although I will occasionally provide examples of other clause-types.

4 proves that, despite their being DPs, FCCs are Case-less. Finally, Section 5 lays out my analysis of this paradox, which is that FCCs make use of proxies to ensure their activation in the structure and vicariously participate in the derivation.

2 Finite complement clauses are DPs

This section shows that, despite the lack of nominal morphology, FCCs can be considered as DPs in Classical Greek. Compelling arguments for this categorization come from their distribution (2.1), the possibility of determining and modifying them in the same fashions as DPs (2.2), agreement (2.3) and association with articles in synchrony and diachrony (2.4).

2.1 Distribution: Alternation and coordination with DPs

First, all Classical Greek clause-taking predicates also take DPs. For example, in (8), the same verb *punthánomai* ‘learn’ appears with a DP (a) and a CP (b). That this holds generally was verified in corpora and dictionaries like Liddell and Scott (1996) for the 105 predicates of the corpus that embed *hóti*-clauses.

(8) a. X. *HG* 2.1.2

Puthómenos [_{DP} **tò sýnthēma**], ...

having.learnt the plot:ACC

‘When (Eteoniceus) learned of the plot, ...’

b. X. *HG* 3.2.11

Puthómenos [_{FCC} **hóti polùs sîtos enên autoîs**], ...

having.learnt that a.lot food was.in them:DAT

‘When he learned that they had a lot of food, ...’

Similarly, DPs and FCCs can be coordinated, as in (9) under the verb *horáō* ‘see’, here appearing in its aorist suppletive stem *id-*.

(9) Pl. R. 496c7

Hikanôs idóntes [_{DP} **tēn manían**] *kai* [_{FCC} **hóti oudeis**

enough having.seen the fury:ACC and that nobody

oudèn hygiés perì tà tōn póleōn práttei]

nothing sane about the the cities:GEN does

‘Because he saw enough the fury (of the multitude) and that nobody does anything sane about the city’s affairs.’

2.2 Determination and modification

Moreover, FCCs share determining and modifying characteristics with DPs. As a baseline, consider the paradigm in (10) centered around the noun *khōrion* ‘village/place’. NPs can be determined with demonstrative determiners like *toûto* ‘this’ in (a-d) and intensifiers like *autó* ‘itself’ in (e). Note that this involves the presence of a definite article. Moreover, there is no adjacency requirements for the demonstrative and the intensifier. In (b), the demonstrative is separated from the NP it bears on by the verb *labôn* ‘taking’. Finally, demonstratives come in three guises: Medial (a-b), proximal (c) and distal (d).⁸

(10) a. Medial demonstrative adjacency, Th. 4.3.3

Toûto tò khōrion.

DEM.MEDIAL the village

‘This village’

b. Medial demonstrative non-adjacency, X. HG 7.5.11

Toûto labôn tò khōrion.

DEM.MEDIAL having.taken the village

‘After taking this village’

c. Proximal demonstrative, X. An. 5.7.15

Ei láboi **tóde** tò khōrion

if they.took DEM.PROX the place

‘If they took the place, ...’

d. Distal demonstrative, Th. 1.87.2

“Anastētō es **ekeîno** tò khōrion”, deixas ti khōrion autoîs.

rise:IMP.3SG to DEM.DIST the place showing some place them:DAT

‘“Let him arise and go to this place”, showing them a certain place.’

e. X. An. 5.5.20

Hê hēmâs edécheto **autò** tò khōrion, taútē eiselthóntes.

where us received itself the village there going.in

‘Going in at a point where the place itself received us.’

The same modifying properties are offered to FCCs, as visible in (11): Adjacent (a) and non-adjacent (b) demonstratives; proximal (c) and distal (d) demonstratives; intensifiers (e).

⁸ All can be adjacent or non-adjacent.

- (11) a. Medial demonstrative adjacency, Antipho 5.32

Oîmai hymâs epístasthai *toûto*

I.think you:ACC know:INF DEM.MEDIAL:ACC.N.SG

[hóti eph' hoîs àn tò pleîston méros tês basánou, pròs toutōn eîsîn hoi basanizómenoi légein hó ti àn ekeínois méll ōsi charieîsthai].

'I think that you know it [that witnesses under torture are biased in favor of those who do most of the torturing; they will say anything likely to gratify them].' (Maidment adapted)

- b. Medial demonstrative non-adjacency, X. *HG* 3.5.15

Toûto khrè eû eidénai **[hóti hē Lakedaimoníōn pleonexía**

DEM.MEDIAL:ACC.N.SG is.necessary well.to.know that the Lac.:GEN greed

polỳ eukatalytōtéra estí tês hymetéras genoménēs archês].

much more.easy.to.overthrow is than yours having.been empire

'It must be well known that the Lacedaemonians' greed is much easier to overthrow than your once empire.'

- c. Proximal demonstrative, Pl. *Plt* 268a5

Toûto mèn episkepsómetha, *tóde* dè ísmen

DEM.MEDIAL:ACC.N.SG PTC we.will.examine DEM.PROX:ACC.N.SG PTC we.know

[hóti boukólō ge oudeîs amfisbētēsei perì toutōn oudenós].

that cowboy:DAT PTC nobody will.contest about among.these nothing

'We will examine this point, but we know (the fact) that with a cowboy nobody will contest any of these.'

- d. Distal demonstrative, X. *Cyr.* 2.4.25

Mémnēso *ekeîno* **[hóti fthánein deî**

recall:IMP.2SG DEM.DIST:ACC.N.SG that be.first must

pefragménous tous pórous prîn kineîstai tèn thēran].

plugged the passages:ACC before move:INF the hunt

'Recall that the passages must be plugged before the hunters move.'

- e. Modification with *autó* 'itself'

S. *Phil.* 1020-1024

Egō algýnomai

I am.suffering

(*toût'*) *aúth'* [hóti zô sÿn kakoís polloís tálas].

this itself:ACC.N.SG that I.live with bad a.lot unfortunate

'As far as I'm concerned, what I am suffering from, is that I live with a lot of bad persons, unfortunate.'

(11)a-d seem to be liable to another analysis, in which the demonstrative is not a determiner but a pronoun which is coreferential with the clause, as in English 'Peter resented *it* that Mary hadn't come to his birthday party'. However, we are dealing here with a demonstrative and not with an expletive pronoun like *it*, which make it suspect that they have the same syntax.⁹ Although space prevents a full comparison of the two analyses,¹⁰ we can concentrate on the predictions that each makes in terms of referentiality. In particular, if the demonstrative determines the clause, together they should act referentially like any DP with a demonstrative. Conversely, if the demonstrative points at the clause, no such behavior should be observed.

To assess this hypothesis, consider (11)a-d again. Note that if the only function of the demonstrative is to point at the following clause, there is no reason to use the same alternation as with DPs. In contrast, if the whole phrase demonstrative+clause is used referentially, the alternation is meaningful. In fact, in discourse, *toûto* and *tóde* are used complementarily, the former to resume the previous move of conversation, the latter to announce the next move. This is the configuration that (c) exhibits. Likewise, in (a) and (b), the content of the whole sequence medial + FCC is used to sum up the previous conversation. Finally, in (d), the distal *ekeîno* +FCC has a memorial function. It serves to advert to facts that are not deemed by the speaker active in the memory of the addressee. It is particularly frequent with general truth. Moreover, if the demonstratives were pointing at the FCC, they would not be available at the same time for another anaphoric or deictical operation.¹¹ This means that FCCs behave like definite DPs in having referential, especially anaphoric and deictic properties.

Finally, the possibility of intensifying is telling. This test was first raised in Davies and Dubinsky (1998), who observe that, in English, subject clauses and subject DPs pattern together in allowing intensification with *x-self* (12) (= their 8 and 11a), which is not the Case with objects. Object DPs can cooccur with *x-self* (13)a), but not object clauses (13)b) (= their 9 and 13a). From this, they draw the conclusion that subject, but not object clauses can be DPs.

(12) a. **The professor** *herself* offered the student sage advice.

b. **That Leslie arrived drunk** *itself* put Kelly in a foul mood.

(13) a. The zookeeper forced **the monkey** *itself* to clean up the cage.

b. Kelly was angry **that Leslie arrived drunk** (**itself*).

⁹ In German, both the pronoun *es* and the demonstrative *das* are possible with an FCC. See Sudhoff (2003; 2016) for the idea that the former is an expletive whereas the second is a determiner.

¹⁰ Faure (2021: 292-308) explores them in detail, along with two other possibilities.

¹¹ More on this in Faure (2019).

Now consider (11)e. In Classical Greek, object clauses allow for intensifiers, the same way DPs do ((10)e). Following Davies and Dubinsky's (1998) rationale, Classical Greek FCCs must be analyzed as DPs.¹²

2.3 Agreement

Another property of DPs is that they carry ϕ -features (person, number, gender). In (14)a and (b), a masculine *émporoi* and a feminine *agélai* noun in the plural trigger plural agreement on the verb and plural and masculine, resp. feminine agreement on the determiner and the predicative adjective (*filósitoi* and *khalepóterai*). In contrast, although neuter nouns do trigger neuter and number agreement on the determiners and predicative adjectives (*éthnē* -> *hupékoa*-c, *pséfisma* -> *iskhurótaton*-d), they do not on the verb, which must be in the singular, which has consequences on the testing of FCCs and their carrying singular or plural number.

(14) a. Masculine plural, X. *Oec.* 20.27

Hoi	émporoi	filósitoí	eisi.
the:NOM.M.PL	merchant:NOM.PL	corn liker:NOM.M.PL	be:IND.PRS.3PL

'Merchants like corn.'

b. Feminine plural, X. *Cyr.* 1.1.2

Khalepóterai	eisin	hai	agélai.
more.difficult:NOM.F.PL	be:IND.PRS.3PL	the:NOM.F.PL	herd:NOM.PL

'Herds are more difficult to manage.'

c. Neuter plural, X. *HG* 6.1.9

pánta	tà	kúklō	éthnē
all:NOM.N.PL	the:NOM.N.PL	around	people:NOM.PL

hupékoa	estin.
submitted:NOM.N.PL	be:IND.PRS.3SG

'All the surrounding peoples are submitted.'

d. Neuter singular, X. *HG* 1.4.3

Tò	Kannōnoû	pséfisma	esti	iskhurótaton.
the:NOM.N.SG	Cannonos:GEN	decree:NOM.SG	be:IND.PRS.3SG	strongest:NOM.N.SG

¹² The reason why English object clauses cannot appear with *x-self* is somewhat a mystery. The intensifiers *autó* and *x-self* are distinct in that the latter is reflexive-based, but not the former. I speculate that the degradation of (13)b) may be due to independent reasons from the categorial status of the clause, linked to the licensing conditions of *x-self* in English. In particular, if the object clause is blocked in a low VP as I claim, it might not reach a position from where it can bind *itself* properly. The exploration of this hypothesis is left for future research.

‘Cannonos’ decree is the strongest.’

In (11), the demonstratives that bear on FCCs are in the neuter. Be that a default gender or agreement with the clause, FCCs thus pattern with neuter nouns and do not trigger agreement on the verb. Therefore, only determiners and modifiers agreement can be used to check for the presence of ϕ -features on FCCs.¹³ In addition to neuter agreement, there is also number agreement in the examples in (11). That this agreement is not default is shown by the alternation with a plural agreement as on *taûta* in (15).¹⁴

(15) Demonstrative plural agreement, X. *HG* 2.3.53

Kaì *taûta* gignôskontes

also DEM.MEDIAL:ACC.N.PL knowing

hótioudèn tò emòn ónoma euexaleiptóteron è tò hymônhekástou.

thatnothing the mine name more.erasable thatthe of.you each:GEN

‘Especially when you know that my name is not easier to erase than that of each of you.’

Consequently, FCCs share one more property with DPs, their carrying ϕ -features. This proves wrong Manzini and Savoia's (2018: 254) proposal that the syntactic behavior of FCCs is due to their missing ϕ -features.¹⁵

2.4 Synchronic and diachronic complementary distribution: Complementizer *h-(óti)* is a D
Finally, if FCCs are nouny, we expect them to share with nouns not only ϕ -features, demonstrative determination, but also the capacity to be endowed with a definite article,¹⁶ contrary to facts: **To hóti...* (‘the that’) never surfaces in Classical Greek, although there are 22,325 tokens of *hóti*.¹⁷ This is surprising for two reasons.

The first reason is that the definite article in Classical Greek can nominalize anything: Finite verbs, e.g., *toû ên te kaì éstai* (lit. ‘the was and will.be’, i.e., ‘the past and the future’); adverbs *toû potè, tò épeita, tò nyn* (lit. ‘the once, the then, the now’, i.e., ‘the past, the future and the present’ (Pl. *Prm.* 152b3-6). Crucially, the only category that the article cannot nominalize is finite declarative complement clause.¹⁸ One way to make sense of this unexpected restriction

¹³ One special configuration, involving a participial, nevertheless shows that there is also agreement between v° and the FCC. It is explored in Section 5.2.

¹⁴ The conditions of this alternation are poorly understood. Hypotheses are explored in Faure (2021: 381-391) along ontological lines (i.e., the type of object that the clause denotes), but more research is needed.

¹⁵ On this, see already Picallo (2002).

¹⁶ Classical Greek does not have an indefinite article.

¹⁷ TLG® (<https://stephanus.tlg.uci.edu>, Project Director: Maria Pantelia), accessed October 22, 2024. Unfortunately, since, in most works, the TLG orthographically does not distinguish *hóti* from the homonymous *wh*-pronoun *hóti*, it is likely that many of the 22,325 tokens are false positives. Be that as it may, the number of complementizers must be substantial.

¹⁸ Indirect interrogative can be nominalized.

is to posit a complementary distribution between *tò* and *hóti*. In fact, (unsurprisingly) definite determination is not recursive, i.e., the definite article and definite DPs cannot be over-determined with another definite article.

The second reason is that the heirs of *hóti*-clauses, namely Modern Greek *óti*-clauses, may be nominalized with the definite article, as repeatedly observed in the literature (e.g., Roussou 1991). See for example (16), from Faure (2023: 82).

- (16) Epiveveóni *to* **óti** **éxis** **teliósi** **ta** **mathímatá** **su**.
 Confirm:3SG the that have:2PL finished the homework yours
 ‘He confirms that you have finished your homework.’

There are important differences, however, between Classical and Modern Greek. Whereas the former limited their usage to factive and quotative/response-stance verbs, the distribution of (*h*)*óti*-clauses is larger in the latter, extending to non-veridical attitude and response-stance predicates. Faure (2023) observed a correlation between the extension of the distribution, the rise of the possibility of nominalization and a third factor, namely the loss of the initial *h*-morpheme. This *h*-morpheme is responsible for the definiteness of a number of items in which it shows up, such as identificational relative clauses (Faure 2021b). It is in complementary distribution with the indefinite/interrogative morpheme *t-/p-*, as shown in (17).

- (17) Relative *h-oû* ‘where’ *h-óthen* ‘whence’ *h-ê* ‘whither’
 Interrogative *p-oû* ‘where?’ *p-óthen* ‘whence?’ *p-ê* ‘whither?’

The possibility of replacing *h-* by *to* in the history of Greek indicates that *h-* played the role of a definite article in Classical Greek, which explains the definite behavior (factive/quotative) of *hóti*-clauses.

2.5 Interim conclusion

In this section, taking the example of *hóti*-clauses, we have seen that Classical Greek FCCs display the trappings of DPs in the language: They are selected by DP-selecting verbs, they can be determined and intensified the same way. Moreover, they carry ϕ -features, as can be seen by their agreement. Finally, their complementizer decomposes into *h+óti*, the first morpheme of which has the semantics and part of the syntax of a definite determiner.

However, this analysis cannot be the whole story. As pointed out in Introduction, *hóti*-clauses do not syntactically behave like other DPs when they are arguments. They have a much more restricted distribution, being limited to the clause-final position, except under very strict (and telling) conditions. This puzzle is examined in the next two sections, starting with their fixed position.

3 Finite complement clauses are *in situ*

In the previous section, I argued that FCCs are DPs. As such, they are expected to be Case-bearers. To be able to explore this prediction in Sections 4 and 5, we must first consider their distribution in the clause, which is the topic of the present section. First, when we compare their distribution with that of *bona fide* DPs, it turns out that it is much more limited. They show a strong preference for the clause-final position (Section 3.1). This feature, which holds crosslinguistically, was variously analyzed in the literature. In Classical Greek, it is due to their embedding deep within vP (Section 3.2).

3.1 Distribution of FCCs in the clause

To assess the distribution of FCCs as DPs, they must first be compared to that of run-of-the-mill DPs. First, consider (18) and the behavior of the DP *tòn stratón* ‘the army’.

- (18) a. Part of a focal domain, Th. 1.64.2

Proségage *tê* *Poteidaíā* ***tòn stratón***.

Led the Potidaea:DAT the army:ACC

‘(Phormion) led the army to Poteidaia.’

- b. Topic, Th. 1.11.1

Tês trofês *aporía* ***tòn stratòn*** *elássō égagon*.

the food:GEN lack:DAT the army:ACC smaller they.led

‘Because of the lack of food, (lit.) the army, they led (it) reduced.’

- c. Preverbal Narrow Focus, Ar. V. 1086

Glaûx gàr ěmôn prîn *mákesthai* ***tòn stratòn*** *diéptato*.

owl for our before fighting the army:ACC had.flown.over

‘For before the battle an owl had flown over the army.’

In Classical Greek, information properties are considered the major factor that rules word order (Dik 1995). In particular, the language distinguishes between the postverbal and preverbal fields and two situations (Matić 2003): When a constituent is focused (Narrow Focus) and when a large portion of the clause is in the focus, typically the whole verb phrase (Focal Domain). (19) schematizes these two configurations.¹⁹

¹⁹ This is an outrageous simplification, but it is sufficient here. The interested reader is referred to the works mentioned in this paragraph and to Bertrand (2010), for a fine-grained mapping between discourse functions and structural slots. Topics, in particular, come in various types (frame-setting, aboutness, etc.).

(19) Classical Greek Word-order configurations

a. Narrow-Focus configuration

Topics > Narrow Focused element > V > rest of the clause

b. Focal-Domain configuration

Topics > $\emptyset$ > [Focal Domain V > rest of the clause]

As can be seen from (18), with the example of the DP ‘the army’, a run-of-the-mill DP can be endowed with any information function and occupy any slot in the structure. This is in contrast to FCCs. Consider (20).

(20) a. Th. 1.144.3

Eidénai chrè hóti anánkē polemeîn.

to.know is.necessary that is.obligatory fight:INF

‘It must be understood that fighting is unavoidable.’

b. X. HG 3.1.8

Katēgórōun autoû hoi sýmmachoi

accused him the allies

hōs epheîē harpázein tô strateúmati toûs phílous.

that he.allowed to.grab the.army the friends

‘The allies accused him of allowing his army to plunder their friends.’

c. Th. 3.52.2

Eirēménōn ên autô ek Lakedaímonos,

being.said:N.SG.ACC was to.him from L.

hópōs (...) mē anádotos eíē hē Plátaia.

that NEG to.be.given.up was the P.

‘Order was given from Lacedaemon that Plataea might be lost.’

As is obvious, every FCCs whatever the complementizer that heads them, whatever their information function, appears clause-finally.

3.2 Clause-finality is *in-situ*-ness

The restriction of FCCs to the clause-final location holds more generally crosslinguistically and was often analyzed as an extraposition of the FCCs, i.e., an attachment to vP (or a higher projection), in a peripheral, adjunct position, *via* Rightward Movement (Stowell 1981 and

Bruening 2018, a.o.–English; Beerman, LeBlanc, and Riemsdijk 1997 (ed.)–for a range of languages). In some works, more steps are posited (Left- or Rightward Movement followed by Remnant vP Movement, Bhatt and Dayal 2007–Hindi; Moulton 2015–English). However, there are alternative positions, which are based on evidence that FCCs behave as if they were within vP (Zwart 1993–Dutch; Haider 2010–German; Sheehan 2010–English; Faure 2021a–Classical Greek), the correct word order being derived *via* post-syntactic, PF extraposition for some authors (Truckenbrodt 1995, Féry 2015 and Moulton 2015–German; Manetta 2012–Hindi; Potsdam 2022–Malagasy).

A wide range of facts were enlisted to test for these various possibilities, including: VP-fronting; Though-movement; Pseudo-clefting; vP-ellipsis; Pseudo-gapping; Relative ordering with adverbials; Conditions A and C of the Binding Theory; Quantifier scope; NPI-licensing; Ā-extraction; A-extraction; Clause-boundedness (Right Roof Constraint); position with respect to verbal clusters in Germanic SOV languages.

It is beyond the scope of the present article to survey all of these criteria. They were not tested for all the languages for which extraposition was hypothesized and several of them were tailored to English or German (such as vP-ellipsis, verbal clusters) and cannot easily be carried over to other languages. Moreover, some of them return contradictory results, such as VP-fronting (Drummond 2009). However, two return conclusive results for Classical Greek: A-extraction and NPI-licensing.

3.2.1 A-extraction

Classical Greek offers examples like (21), in which a constituent of the embedded clause, its subject, here *toútous*, surfaces in the matrix, where it is marked in the accusative by the matrix verb and not in the nominative, as expected from the subject of a finite verb.

(21) D. 43.23

Aisthésesthe *toútous*

you.will.fell these:ACC.M.PL

hóti ~~hoútoi~~ eisi bíairoi kai aselgeís ánthrōpoi.

that they:NOM.M.PL are violent and reckless men:NOM.M.PL

‘You will gather that these men are violent and reckless.’

This phenomenon is traditionally called *Prolepsis* and has raised a debate about its being movement (Kühner and Gerth 1904: 577, §600.4)²⁰ or not (Milner 1980). The evidence is contradictory, so that two phenomena probably cohabit in Classical Greek, as shown in Faure (To appear).²¹ However, at least one of them involves derivation *via* movement. First, positing movement out of the embedded clause endows the DP with a θ -role, of which it would be

²⁰ “Das Subjekt des Nebensatzes in den Hauptsatz herübergangen und hier zum Objekte gemacht wird” ‘The subject of the subordinate clause is raised to the matrix, where it is made object.’

²¹ This is also true for other languages and crosslinguistically. See Lohninger, Kovač, and Wurmbrand (2022) for an overview.

deprived if it were generated in the matrix and coreferential with a null pronoun in the subordinate. Second, a movement view accounts for Cases of split DPs, as illustrated by (22).²²

(22) D. 19.203

Epideîxai kaì hóti pseúsetai
 to.show both that he.will.lie
 kaì tēn dikaían hētis estin ~~hē—dikaía~~ apología.
 and the just:ACC what is the just:NOM defense:NOM
 ‘to show both that he is going to lie and what a just defense is.’

In this example, the DP *hē dikaía apología* ‘the honest defense’ is generated in the embedded clause as a subject and is then split into two parts, the article and the adjective raising to the matrix, while the noun is stranded within the clause. Crucially, the raised part receives the accusative case, whereas the stranded noun retains the nominative, the case allotted to subjects of finite clauses.²³ Note that this looks like a movement of a non-constituent, but this severing of the group article+adjective from its noun is freely available in Classical Greek (Devine and Stephens 2000: 45-48). (23) is an example of such an operation, within the same clause.

(23) Antipho 1.27

Tēs dikaiotátēs àn túkhoi ~~tēs—dikaiotátēs~~ timōrías.
 the justest:GEN PTC she.would.obtain the justest:GEN punishment:GEN
 ‘She would obtain the justest punishment.’

This is why I claim that at least some instances of Prolepses are Cases of Hyperraising to Object. Wherever the target object position is (in a VP-shell, in Spec,vP), it must c-command the FCC to be able to probe the raised DP out of it. Hence, FCCs must be low within the vP-domain. A side conclusion is that Case in Classical Greek cannot be “dependent”, in Marantz's (1991) sense that (structural) Case-assignment depends on the case-feature that another DP in the

²² More on this in Faure (2018a, b), where the analysis is couched within the frame of Phase Theory (Chomsky 2000, see also below section xxx). Problems for movement theories like improper movement are also addressed there.

²³ Note that in (21) and (22), I noted the erased copy on the embedded clause as carrying a nominative Case. In (21), the accuracy of this analysis is visible from the agreement in the nominative of predicative NP *báioi kai aselgeís ánthrōpoi* ‘violent and reckless men’. In Faure (2018b), I suggest that the raised constituent keeps its nominative Case, but that it is overwritten by the accusative according to a hierarchy of Cases. It would take us too far astray to make the full proof, but the process would be comparable to what Richards (2013) observes in Lardil.

same phase²⁴ has or does not have. In particular, we observe that Case-assignment to Classical Greek DPs does not hinge upon the Case of another DP but upon the agreement with a functional head that probes for it.

3.2.2 NPI-licensing

Another operation that supposes that the FCC is low within vP is the licensing of a Negative Polarity Item (NPI). As a baseline, consider the pair of examples in (24) (b—from Chierchia 2013, a—inspired by Collins and Postal 2014). In (24)b, the verb *doubt* licenses the NPI *any (policeman)*.²⁵ This licensing is blocked if the FCC is withdrawn from the verb's c-command domain, say, via Topicalization, as in (a). (a) also shows that reconstruction of the clause in its base-position at LF does not rescue the structure. This paradigm proves that NPI-licensing takes place before Spell-out. Moreover, granted that the English verb remains within vP (Emonds 1978), the FCC that contains the NPI cannot be higher than the verb (adjoined to vP as in (25)(b) and must be lower, within vP as in (25)(a).

(24) NPI-licensing

- a. *[_{FCC} That we need **any policeman** in that corner], I [_{VP} *doubt* ~~that~~...]].
 b. I [_{VP} *doubt* [_{FCC} that we need **any policeman** in that corner]].

(25) a. [_{VP} ... licenser ... [_{FCC} ... NPI ...]]

- b. *[_{VP} [_{VP} ... licenser ...] [_{FCC} ... NPI ...]]

This demonstration extends to Classical Greek. In Classical Greek most finite verbs remain in vP as well²⁶ and licensing of NPIs can take place crossclausally. For example, consider *pópotē* 'ever', which appears in (26) and (27).

(26) Pl. Ap. 33d3

<i>eíte</i>	<i>tinès</i>	<i>autôn</i>	<i>presbúteroi</i>	<i>genómenoi</i>	<i>égnōsan</i>
if	some	of.them	older	having.become	understood
<i>hóti</i>	<i>néois</i>	<i>oûsin</i>	<i>autoîs</i>	<i>egò kakòn pópoté ti</i>	<i>suneboúleusa</i>
that	young	being	them:DAT	I bad ever smthg	suggested

'If some of them after growing older understood that, when they were young, I had ever suggested something wrong to them, ...

²⁴ Adopting Baker and Vinokurova's 2010 update. See also Bobaljik (2008), McFadden (2009), and Baker (2015, esp. the formulation p. 48-49). Baker (2015: Chap. 2) shows that both dependent and independent procedures of Case assignment exist across languages.

²⁵ For a theory of this type of crossclausal licensing, see Progovac (1994).

²⁶ For example, in 97% of Faure's (2021a: 34-37) corpus, the finite verb appears after the adverb *pollákis* 'often'. Note that Matticchio and Sanfelici (2021) found more nuanced data and argue that the position of Classical Greek verbs is not fixed and can be either in T° or in v°.

(27) D. 45.6

Hubristheîs hōs *ouk oîd'* **eí tis** *pópot'* állos anthrōpōn apéein.

mistreated how NEG I.know if someone ever else men:GEN I.left

‘Treated as badly as I doubt if any other man has ever been, I left.’

In (26), *pópotē* ‘ever’ is licensed in the *hóti*-clause because its matrix verb is itself embedded in an antecedent of conditional (*eíte*). Note, however, that this licenser is higher the vP itself and does not preclude an adjunction-to-vP analysis of the FCC. *Pópotē* is also found in an *ei* ‘if’ clause in (27), the licenser is the negative *ouk*, which is situated right above the verb [_{negP} *ouk* [_{vP} *oîda*]]. In order for *pópotē* to be licensed, its host clause must be deep inside vP, probably in its merge position.

3.3 Interim conclusion

In this section, we have seen that Classical Greek FCCs strictly appear clause-finally and it was shown that they structurally are low within vP. Since Classical Greek is liberal in terms of placement of DPs, this limited placement is unexpected under the view that FCCs are DPs, following the conclusions of the previous section. Although this finding might challenge this categorization, it does not, once we take into account another feature, namely Case-marking.

4 Finite complement clauses are Case-less

Even though a number of arguments point towards FCCs being DPs (Section 2), they do not distribute like DPs in the clause (Section 3). Another, puzzling fact is that FCCs do not display case-marking. This absence is all the more surprising, given that, in Classical Greek, any DP is morphologically case-marked. In Classical Greek, case morphology is the exact reflex of abstract Case (cf. Legate 2008). In particular, there is morphological distinction between structural, lexical and inherent Cases in the sense of Woolford (2006) (Faure 2010: 201-204). The nominals that are objects to propositional attitude verbs are accordingly marked in the accusative, as we saw in (8)a (rarely in the genitive).

In the present section, we will see that this absence of overt case-marking does not disguise covert case-marking and that Classical Greek FCCs are actually Case-less. This can first be inferred from usual tests: They cannot appear in subject position or with preposition, in contrast to non-finite clause-types (Section 4.1). This first approach will be buttressed by two tests that are rarely used in this perspective: FCCs’ reluctance to appear in object positions (to be defined) and their derivational behavior (Section 4.2). The latter is key to understanding the link between *in-situ*-ness and Case-less-ness.

4.1 Resistance to subject position and prepositions

It has long been observed that FCCs are reluctant to appear in subject position and after prepositions (Stowell 1981), although there are exceptions, even in English (Davies and Dubinsky 1998; Hartman 2012). Classical Greek FCCs are not different. They are never found after prepositions.

As far as the subject position is concerned, given the flexible word order of Classical Greek, which was presented in 3.1, this criterion might not be relevant. However, subjects still have a default, preverbal position, which can be argued to be Spec,TP (Faure 2021a: 30-34) above the

Narrow Focus and below the Topic positions, whence the scheme: Topics > Subject > Narrow Focus > V. The subject location does not seem to be available to FCCs. For example, consider *hóti*-clauses with *aggéllō* ‘announce’ in the passive in Thucydides. DP subjects can appear both pre- (28) and postverbally (29). In contrast, subject *hóti*-clauses steadily appear postverbally, be they in the same configuration as (28) (after an adverbial participial clause = (30)) or (29) (after an indirect object = (31)).

(28) Th. 8.11.3

Hrmēménōn autōn [DP **tà** **perì tēn en tō Speiraíō**
being.launched:GEN them:GEN the:ACC.N.PL about the in the Sp.
tōn neon katafugēn] ēggélthē.
the ships:GEN chasing was.announced

‘lit. While they were about to go, the event of the ships being chased in Speiraios was announced.’

(29) Th. 2.6.2

Tois Athēnaíois ēggélthē euthùs
The Ath.DAT was.announced right.away
[DP **tà perì the Pl. gegenēména]**
the about the Plataeans events

‘The events taking place in Plataea were announced to the Athenians.’

(30) Th. 5.10.2

Perì tò hieron tēs Athēnās thuoménou (...) *aggélletai* (...)
about the shrine of.the Ath. being.sacrificed:GEN is.announced:PRS.IND.3SG
[FCC **hóti hē stratìà hápasa fanerà tōn polemíōn en tē pólei].**
that the army whole visible the enemy:GEN.PL in the city:DAT

‘While a sacrifice was going in the temple of Athena, it is announced that the whole enemy army is visible in the city.’

(31) Th. 6.45.1

Tois Surakósiois (...) *ēggélleto*
the Syracusians:DAT was.announced:PST.IND.3SG
[FCC **hóti en Regíō hai nêés eisi].**
that in Rhegium the ships are

‘It was announced to the Syracusians that the ships were in Rhegium.’

FCCs do appear preverbally in the corpus, although extremely rarely (3/33 subject *hóti*-clauses). All three cases are like (32),²⁷ where the subject clause is accompanied by the postpositive *mén* particle, which indicates that it is a topic. Although in principle it is imaginable that the particle is attached to the subject, the positions Topic and Subject are distinct in the scheme of the Classical Greek clause as seen above and there are prosodic signs of detachment in initial position that go along with the presence of the *mén* particle, such as the possibility of inserting a vocative (*ô boulé* in the example).

(32) Initial subject clause, Lys. 3.15

Hóti mén toínun hoûtos ên ho adikésas, ô boulé, (...)
 that PTC then this.man was the injust PTC council:voc
 hypò tôn paragenoménōn memartýrētai hymîn.
 by the orators was.witnessed by.you

‘That this man was the culprit, Council, and conspired against you and that I did not against him was witnessed to you by those who were present.’

Finally, the contrast is revealing between FCCs and non-finite (infinitival) clauses, of which (33) and (34) are two examples.

(33) Infinitival clause with a preposition, X. HG 1.3.19-20

Dià taût’ toûs polemioús éphē eisésthai
 because.of that the enemy:ACC he.said will.be:INF
 ou dià [tò miseîn Lakedaimoníous].
 not because.of the hate:INF Lac.:ACC

‘He had for this reason admitted the enemy, not for the sake of money nor out of hatred to the Lacedaemonians.’

Lit. ‘Because of the fact that he hates the Lacedaemonians’

²⁷ The other two examples are Lysias 3.40 and 13.72.

(34) Infinitival clause in subject position, Th. 1.142.6

Tò tês thalássēs epistémonas genésthai

the the sea specialists become:INF

ou rhāidíōs autoîs prosgenéseai.

NEG easily to.them will.be.added

‘They won’t acquire easily knowledge about the sea.’

Infinitival clauses can be endowed with a Case-bearer definite article. In these conditions, such clauses do appear after preposition (33) and in subject position (34).²⁸ This last data point confirms that Case is the licenser at play.

4.2 Resistance to $\bar{A}$ -movement in Narrow Syntax

Another test confirms the role of Case, namely the impossibility of moving around, especially to $\bar{A}$ -positions, when Case-less. I assume that the verb phrase is a complex domain made of several projections, in which a root projection (noted between two $\ddagger$) is surmounted by a vP projection. The root is categorized as verbal by v° , as in the current version of Distributed Morphology (Marantz 2007). In $\ddagger P$, the DP object is merged with its root selector and receives a θ -role. For example, the relevant steps of the derivation of (18)a (repeated here) can be represented as in Figure 1. The verb root has at least three allomorphs (*(pros)ag-*, *agōg-*, *agag-*). For ease of exposition, I use the shortest one here. Case-assignment does not take place in a local configuration between v° and the direct object DP but remotely (see the technical discussion in Section 5).

(18) a. Part of a focal domain, Th. 1.64.2

Proségage tē Poteidaíā tòn stratón.

Led the Potidaea:DAT the army:ACC

‘(Phormion) led the army to Poteidaia.’

²⁸ Interestingly, the same possibilities exist in Modern Greek when *óti*-clauses are nominalized with a definite article (Roussou 1991).

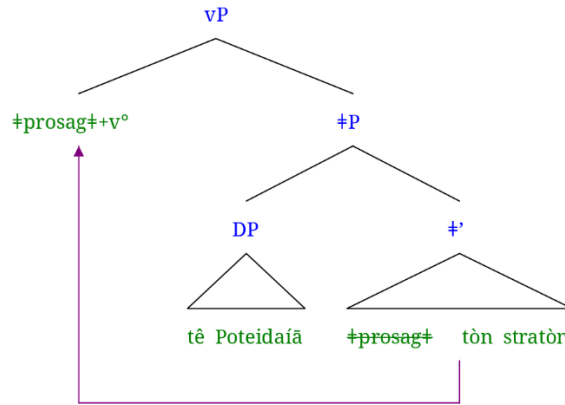


Figure 1: Representation of (18)a

We saw in Section 3 that FCCs are not at the periphery of the verb phrase, so not in Spec,vP, meaning that they must be in the complement of $\bar{\alpha}$, i.e., *in situ*, where they were merged. This also explains why they appear clause-finally. Here I claim that they do not get Case there, which accounts for why they are not able to move around. Apart from the freedom that we observed in (18) for case-marked DPs and that FCCs do not have, there are two sets of facts that support this analysis. First, in languages where case-marking does appear on FCCs, the clause-finality requirement is cancelled. Consider for example the Korean example in (5). The *kes*-clause is marked in the accusative (-*ul*) and appears preverbally.²⁹

(5) Korean (Bogal-Allbritten and Moulton 2018: 220, glosses as in original)

Na-nun [ney-ka swukecey-lul ta ha-yess-ta-nun kes-**ul**] mit-e.

I-TOP you-NOM homework-ACC all do-PST-DEC-ADN kes-ACC believe-DEC

‘I believe (the claim) that you finished your homework.’

Second, Case-assignment is a condition to be able to leave vP, arguably by movement *via* the escape hatch that Spec,vP constitutes according to Phase Theory (Chomsky 2000). Consider first the possibilities that are open to run-of-the-mill DPs in (35) and (36).

²⁹ Other licensors exist, such as nominalizers and ‘say’-related complementizers. Nominalizers like definite determiners were introduced in the previous section for Classical Greek infinitival clauses (see also the Modern Greek example (16)). ‘Say’-related Cs can be illustrated with Bangla *bole*-clauses, which appear preverbally, whilst non-‘say’-C (*je*)-clauses are clause-final. See the following example, adapted from Bayer (1995: 59), and meaning ‘The boy has heard that his father will come’ (glosses as in original, except for *bole* and *je*):

(i) chele-Ta (or baba aS-be **bole**) Sune-che (**je** or baba aS-be)
 boy -[cl] his father come-[fut] [‘saying’-comp] hear-[past] C his father come-[fut]

(35) Eur. *Ba.* 173-174

Ítō tis, eisággelle *Teiresías* **hóti** ~~*Teiresías*~~ **zēteî** **nin**
 come:IMP.3SG someone announce:IMP.2SG T.NOM that looks.for him

‘May someone come. Announce that Tiresias is looking for him.’

(36) X. *Mem.* 1.3.8

Kritóboulon pote *tón* *Krítōnos* puthómenos

K.:ACC PTC the:ACC Kr.:GEN having.learnt

hóti ~~*K. tón Kr.*~~ **eflēse** **tòn** **Alkibiádou** **huiòn** **kalòn** **ónta**.

that loved the Alc.:GEN son beautiful being

‘Having learnt one day that Critoboulos, the son of Criton, was in love with Alcibiades’ son, who was beautiful.’

In (35), the embedded subject *Teiresías* is topicalized right in the front of the complementizer. It retains the nominative case that was assigned to it in the subordinate. Conversely, in (36), *Kritóboulon* is extracted out of the embedded clause past the matrix verb and topicalized within the matrix in a left peripheral position. Note that *Kritóboulon* is in the accusative, the case assigned by the matrix verb. This is a case of *Prolepsis*, an operation that was introduced in Section 3.2.1 and used there as a test to spot the location of the FCCs. Descriptively, accusative marking by the matrix verb is what licenses the movement of the DP to the matrix and the upper field in (36). In (35), the DP does not receive Case in the matrix and thus cannot escape the left periphery of the embedded clause.

As noted at the beginning of this section, Case is not assigned in Spec,vP, but probably long-distantly by v°. However, the establishment of a relation between v° and the case-marked object allows it to pass through Spec,v°. Crucially, it only passes and never ends up its travel there. All instances of preverbal DPs are in a discourse position and not merely in Spec,v°, as visible from (18)b, c, in which the preverbal DP *tòn stratón* ‘the army’ is in a Topic and Focus position, respectively.

(18) b. Topic, Th. 1.11.1

Tês trofês aporía **tòn stratòn** elássō *égagon*.

the food:GEN lack:DAT the army:ACC smaller they.led

‘Because of the lack of food, (lit.) the army, they led (it) reduced’

c. Preverbal Narrow Focus, Ar. V. 1086

Glaûx gàr êmôn prìn mákesthai **tòn stratòn** *diéptato*.

owl for our before fighting the army:ACC had.flown.over

‘For before the battle an owl had flown over the army.’

This is in line with what Kayne (2000) found based on agreement patterns on French participles. Consider the following three examples, inspired by his p. 25-26 and modernized.³⁰ We observe that object DPs need not be in Spec,vP to receive Case and be licensed (a). Furthermore, feminine plural agreement is only triggered on the participle when the direct object (*les, que*) is fronted in a preverbal position (b-c). Finally, the object cannot be left in Spec,vP (d). This paradigm is an indication that preverbal objects pass through Spec,vP (agreement), but cannot end up there. Kayne attributes the latter fact to Spec,vP (his AgrP) not being a Case-position.

(37) French, agreement with a preverbal object

- a. Paul a [_{VP} repaint(*es)+v° [_{VP} les chaises]].
Paul has repainted.F.PL the chairs
- b. Paul les_i a [_{VP} e_i repaint(es)+v° [_{VP} e_i]].
Paul them has repainted.F.PL
- c. Les chaises_i que_i Paul a [_{VP} e_i repaint(es)+v° [_{VP} e_i]].
the chairs that Paul has repainted.F.PL
- d. *Paul a [_{VP} les chaises repaint(es)+v° [_{VP} e_i]].
Paul has the chairs repainted.F.PL

Likewise, the case of Prolepsis (36) is captured in Faure (2018a, b) in the frame of Phase Theory (Chomsky 2000): Spec,vP serves as an escape hatch for the case-marked DP *Kritóboulon* to move out of the vP to reach the higher Topic position. This analysis explains why a DP that is not case-marked in the matrix such as *Teiresías* in (35) is stranded deep within vP (in fact in the embedded CP): It never accesses the escape hatch. Crucially, casting the analysis of Kayne's French data and of Classical Greek Prolepsis within the frame of Phase Theory relaxes the locality requirement that Kayne posited. The stop of the DPs before the verb is not due to agreement operating locally (in a Spec-Head configuration) but to the necessity to halt in the escape hatch to leave vP.

Finally returning to FCCs, we have seen a lot of evidence that they do not receive Case. This absence squares well with the fact that they do not access information/discourse positions, such as the preverbal slots described for (18)b-c and reserved for Narrow Focus or continuous Topics or clause-initial frame-setting topical positions. The only (and rare) exceptions are telling. They invariably look like (38):

³⁰ Kayne's article dates back to 1989, so it does not include Phases and vP. Instead of vP, he posits an Agr(eement) Phrase. We kept his *e* notation for empty categories.

(38) Lysias 9.8

Hóti *mèn* oún **afeithēn** **upó** **tōn** **tamiōn**, epístasthe ~~hóti~~:

that PTC PTC I.was.let.off by the Treasury.clerks know:2PL

prosēkein *dē* hēgoúmenos kai dià taútēn tēn apódeixin apēllákhthai toû enklématos, éti pleíonas kai nómous kai állas dikaiōseis paraskhēsomai.

‘Well, that I was let off by the Treasury clerks, you now know. But although I consider that merely on the strength of this demonstration I ought to stand cleared of the impeachment, I will put in a yet stronger array both of laws and of other justifications.’

In this example from the corpus *Lysiacum*, an object *hóti*-clause is fronted to the left field of its matrix and endowed with the topic particle *mén*. The function of this particle is not just to mark the constituent it applies to as topic, but also as contrastive (Goldstein 2016: Chap. 5). It functions in pair with the particle *dē* that is found in the next sentence: The defendant claims that he is innocent because the clerks released him, but also because the law says so. It is a Case of additive contrast (‘not only... but also’, Krifka 1999). It seems that contrast is here what allows FCCs to move, despite their incapacity to find a phasal way out of vP. I propose that this infringement is due to the fact that contrastive movements are post-syntactic and thus do not abide by the rules of Narrow Syntax, such as phasality. In that, I follow Horvath (2010).

4.3 Interim summary

In this section, we have seen that four data points indicate that FCCs are impervious to case-marking. They do not distribute like case-marked DPs in Classical Greek (they are neither found in subject position nor after preposition) and cannot access the escape hatch out of the vP Phase, which limits their derivational possibility to post-syntactic operations only, like contrastive movements. FCCs being DPs, this fact is unexpected under the accounts for which DPs are generated case-marked (Chomsky 1995; Caha 2009) or must go through a Case filter (Chomsky 1981) or be active (Chomsky 2000), or at least receive default Case (Schütze 2001; Caha 2023). This discrepancy between the categorial status and the absence of case-marking is addressed in the next section.

5 Proposal: Finite complement clauses are vicariously active

We uncovered that Classical Greek FCCs are 1) DPs and 2) Case-less, based on the properties summarized in Table 1. Moreover, I argued that they are 3) *in-situ*, i.e., in a projection low within vP.

	Properties	DP	FCCs
DPhood	Selected by same predicates	✓	✓
	Coordination	✓	✓
	Determination and modification	✓	✓
	Deictic/anaphoric	✓	✓
	Φ/number agreement	✓	✓
	D/C compl. distribution	✓	✓
Case-less-ness	Subject position	✓	✗
	Complement to P	✓	✗
	A-movement	✓	✗
	Ā-movement ³¹	✓	✗

Table 1: Comparison of the syntactic properties of DPs and FCCs

The present section is devoted to capture the combination of these three properties and addressing a number of problems that their cohabitation raises under certain assumptions ruling DPhood and argumenthood.

- Visibility Condition: Being a DP submits FCCs to the *Visibility Condition*, which requires a DP to be Case-marked or to belong to a chain with a Case-marked element, for it to receive a θ -role (Chomsky 1986a,b);
- Activation Condition: The carrying of a Case-feature conditions *Activation* of a DP and its availability for syntactic operations such as agreement with and ϕ -feature valuation of functional heads like T° or v° (as in, e.g., Chomsky 2000).

In the present section, I will review the various hypotheses to check which one fares best with respect to these issues. I will start with Rouveret’s hypothesis that FCCs are inherently visible (5.1), which I will revise. Then I will consider long-distance ϕ -feature valuation (5.2), and the various versions of Agree that can account for it in the case of FCCs (5.3). Finally, I will argue for its depending of a vicarious activation of the FCC via proxies (5.4).

5.1 DPhood, Inherent Visibility and Case-less-ness

Examining non-finite clauses, Rouveret (2023) observes an important difference with FCCs. Simplifying a lot, crosslinguistically, non-finite clauses share the property of being deficient, in that they are tense-dependent or do not have an independent subject or are not properly labeled. This deficiency makes them items that are active in the derivation in search for compensation. In contrast, “featurally complete clauses are autonomous domains that are inherently visible for θ -assignment” (p. 344). Rouveret adopts Chomsky’s (2008) idea that T inherits its feature from the Phase head C/Fin and thus that completeness can be achieved in various ways thanks to mechanisms that apply to T and/or to C. These mechanisms include labeling by a strong T or subject movement to TP and labeling as $\langle \phi, \phi \rangle$, or the presence of overt complementizers. According to Rouveret, an important effect of this Inherent Visibility is that FCCs need not participate in the derivation, which results in their being extraposed, although their feature completeness makes them able to be assigned a θ -role.

³¹ Except when motivated by contrast.

Returning to Classical Greek, we have observed that FCCs do not participate in the derivation but are involved in ϕ -agreement operations. These two properties accord with Rouveret's observations, but not the third one, namely that they are *in situ*, not extraposed. To capture the latter fact along the former two, their categorization as DPs may be sufficient. Conceptually, keeping Rouveret's fundamental idea, DPs are labeled objects, featurally complete and as such inherently visible, in contrast to non-finite structures. This notion of DPs aligns with their being phases. Furthermore, I adopt the widespread idea that Case is not an intrinsic feature of DPs but is contributed by an additional projection (Bittner and Hale 1996; Weerman and Neeleman 1999, a.o.) or several of them (as in Nanosyntax, Caha 2009), in which DPs are inserted. This projection can be viewed as an extended projection of DPs in Grimshaw's (2005) sense. This insertion in a projection that carries a feature that needs checking and valuation creates an active syntactic object.

However, unlike θ -roles, there is no principled reasons why Case would be a natural feature of DPs and I claim that FCCs are instances of bare DPs, with no CaseP above them. As DPs, FCCs are fit for being selected by a verb that subcategorizes for DPs and for playing an argumental/ θ -role in a low VP-shell, where they stay. As DPs, they are also endowed with ϕ -features and able to value the features of a functional head that enter in relation with them. Their inactivity in the derivation is due to their being Case-less.

5.2 Long-Distance agreement: Φ -feature valuation

However, the inactivity of FCCs is problematic for another reason. We observed in Section 2.3, that FCCs agree with other elements in ϕ -features. We pointed out that determiners and modifiers agree with them in number. Moreover, there are signs that they also agree with functional heads v° and T° . Consider (39).

(39) Th. 1.116.3

esaggelthéntōn	hóti Foinissai nêes	ep'	autoùs pléousin.
announce:PTCP.AOR.PASS.GEN.N.PL	that Ph.	ships:NOM towards them	navigate:3PL
lit. 'having been announced that Phoenician ships were navigating towards them.'			

(39) exhibits a genitive absolute. A genitive absolute is an adverbial clause centered around a predication whose predicate is a participle and the subject a DP. Both participle and DP are marked in the genitive, which precisely indicates the adverbial meaning of the whole structure. In the example, the subject is an FCC (which as expected appears clause-finally, *in situ*) and the participle is in the passive. Crucially, the verb agrees in the (neuter) plural with the clause.³² This structure is important, since it shows that agreement is not just local between a determiner that bears on the clause and the clause itself (as in (11)a, d), but can also take place

³² For an example of agreement in the singular, see *eirēménōn* in (20)c).

remotely between the clause and the verb in T° or, likelier here, in v° , a process that we also observed in Section 4 for case-marked DPs.³³

The question arises as to how this long-distance agreement between v° and T° and the FCC can take place. Although remote agreement is not problematic *per se*, it requires the DP goal to be active, i.e., to have a Case-feature to value, as observed in Chomsky (2000). However, recall that we saw plenty of evidence that FCCs are case-less. This issue is addressed in the next section.

5.3 Which Agree?

Let us take stock. From what we observed previously, we know that v° and its object DP/FCC need not be in a local configuration with v° or T° to value their ϕ - and Case-features but take place long-distantly. Moreover, we need Agree to go both ways since DPs probe upward for Case-feature-checking and v° probes downward for ϕ -feature-checking. Alternatively, one might want to maintain a single-direction Agree and reanalyze the seemingly opposite-direction process. All these options were considered in the literature Downward Agree only (e.g., Bošković 2007), Upward Agree only (e.g., Bjorkman and Zeijlstra 2019), both ways (e.g., Baker 2008).

Let us start and consider these options in turn for both run-of-the-mill DPs and FCC-DPs.

In a Downward-Agree-only approach, v° will probe into its c-command domain and agree with the DP in ϕ -features. However, the DP won't be able to probe upward and check its Case-feature. For Case-checking to happen, the DP would have to go to the specifier of v° and probe downward in an EPP-manner. This, however, does not take place, unless we posit a covert movement of which we have no signs. So, we cannot retain a Downward-Agree-only approach to account for our Classical Greek data.

In an Upward-Agree-only approach, everything starts with the DP. It is in need for Case-valuation. Thus, it probes upward, finds v° and gets its feature valued. Moreover, an accessibility relation is established, as defined in (40).

(40) Accessibility relation (Bjorkman and Zeijlstra 2019: 536)

α and β are accessible to each other iff an uninterpretable feature (uF) on β has been checked (via Upward Agree) by a corresponding interpretable feature (iF) on α .

In our situation, the uF is the case-feature on the DP and the DP and v° become accessible to each other. v° then needs to check and value its uninterpretable ϕ -features. Bjorkman and Zeijlstra (2019) allow this valuation to take place downward, as we need here, but only if the feature is first checked upward against a defective item. Put otherwise, the valuation takes place vicariously. What we are missing here to adopt this approach, is the defective item required for the first local (checking) step. Bjorkman and Zeijlstra (2019: 563-564)

³³ Object agreement between an FCC and v° is also (optionally) observed in Bantu languages (Zulu—Halpert 2015; Ndebele—Pietraszko 2019), although some of them, like Logoori, resist it (Gluckman 2021). Gluckman attributes the latter fact to Logoori's FCCs not being DPs, despite their ϕ -agreeing with T° .

acknowledge this problem and leave it for future research, although they sketch a few hypotheses.

Summarizing, whatever unidirectional approach we consider, for FCC-DPs to partake in the ϕ -Agree processes, they need to be active or accessible, which they are not, given the lack of case-features and of vicarious items. Moreover, both Downward- and Upward-only approaches necessitate a locality that we do not observe in our data, where everything takes place long-distantly. Instead, we need a more liberal theory and to relax some of the principles that we discussed in one of two ways: (1) Dispensing with Case-activation; (2) Allowing for bidirectional Agree.

(1) boils down to allowing for non-active or inaccessible items to value features. But this move overgenerates. For instance, in (41), a DP in the (animate) dative plural (italicized) intervenes between the verb in the singular. This DP is not active, since its case is inherent, say, because it was attributed in an applicative projection. If we dispense with Case-activation, since it is the closest DP, a Relativized Minimality configuration is created (Rizzi 1990) and it will value the verb's ϕ -features, contrary to facts (the verb is in the singular, the DP is in the plural).

(41) X. HG 1.1.27

Ēggélthē	<i>toîs</i>	<i>tôn</i>	<i>Surakosiôn</i>	<i>stratēgoîs</i>	oíkothen
was.announced:3SG	the	the	S.:GEN.PL	strategos:DAT	from.home
hóti	feúgoien	hupò	toû	démou.	
that	they.left	from	the	people:GEN	

‘It was announced to the Syracusan strategoi that they had been banished from home by the Democrats.’

(2), bidirectional Agree, is massively entertained in the literature, which assumes that Case is checked parasitically on the operation of ϕ -checking (Chomsky 1995) or opens the possibility that checking and valuation take place in both directions, together or independently (Baker 2008; Bárány and van der Wal 2022), freely between interpretable and uninterpretable features (Pesetsky and Torrego 2007). Overgeneration is also possible. However, even if controlled as it is in the cited works, v° still won't be able to meet FCCs, since they are not active, being Case-less, a problem which remains unsolved and which I tackle in the next section.

5.4 Vicarious activation and null expletives

Instead of dispensing with activation, we must consider that FCCs are active but not in a way that allows them to move around in the derivation. Put otherwise, their activation is different from Case-bearing. That they are active is not only visible in their valuing ϕ -features (see the discussion around (39)), but also in their ability to send over items that are extracted out of them, as in Prolepsis (36), *wh*-extraction (42), and demonstrative severing (15).

Recall that Prolepsis involves a first step in which a nominative DP is extracted out of the FCC before undergoing Case overwriting from the matrix verb. Thus, a proleptic DP carries two

case-features.³⁴ Moreover, Prolepsis is (at least) the topic of the FCC (Panhuis 1984; Faure 2018a). Ignoring what happens within the FCC, I propose the following derivation:³⁵

1. The proleptic DP agrees with the topical complementizer to check its Topic feature; Topic being a criterial position, it must move there (Rizzi 1997);
2. The proleptic DP is active and, I assume, makes its host structure, the whole FCC, active;
3. v° probes into its c-command domain and finds the FCC, with which it agrees;
4. The ϕ -features on v° are valued by the FCC ϕ -features and v° assigns Case to the clause;
5. Case valuation on the FCC only applies to the proleptic DP.
6. Furthermore, the proleptic DP can (cf. ex. (36)) but need not (cf. ex. (21), (22)) be made the topic of the matrix. When it is, it agrees with a topical head in the matrix and moves up to its specifier *via* Spec, v° .

Wh-extraction such as in (42) can be analyzed the same way. *Wh*-extraction of *hó* out of *hóti*-clauses is restricted to *hós*-relativization, which is similar to a stepwise topicalization (Faure 2021b).

(42) Arist. *MM* 1.1.16

epì toûto hò àn boulómetha deîxai ~~hè~~ hóti ~~hè~~ agathón.
to this what:N.SG.ACC PTC we.want:SBJV show:INF that good.

‘(this description applies) to what (lit. that which) we want to show that it is good.’

Demonstrative severing as in (15) provides the same result, although in a slightly different way. The demonstrative is generated on top of the FCC, say, in its specifier, and is part of the DP that the FCC constitutes. Moreover, it is active because of its uninterpretable Case-feature. Consequently, the same trade-off Case against ϕ -features takes place. As in Prolepsis Cases, the demonstrative can (cf. ex. (11)b, d; (15)) but need not (cf. ex. (11)a, d, e) move further up. In the latter case, an information-driven operation occurs. In particular, the clause is focalized. This additional feature triggers attraction to the preverbal (low IP-area), criterial focus projection.³⁶ As in Prolepsis, reaching this projection implies leaving the vP Phase *via* its escape hatch Spec, vP , which we saw is available only to case-marked DPs. Thus, the FCC, despite its activation because of its focus feature cannot undergo this operation. However, the demonstrative in its specifier can and goes there as a proxy.

³⁴ Investigating what makes this possible is beyond this article’s scope.

³⁵ Compare with Bjorkman and Zeijlstra (2019: 554-558) and their discussion of a topic-dependent long-distance agreement in Tsez, as well as literature on the same process in other languages. For their Upward Agree account to go through, they have to posit an invisible case-feature on the complementizer as well as an unvalued interpretable feature.

³⁶ On this, see Dik (1995) and for the low IP-area as a place for information projections across languages, Jayaseelan (1999); Belletti (2004).

This indirect activation of FCCs by Case-bearing elements in their above projection explains the agreement that we observe between v° and the FCC in certain cases, despite the Caselessness of FCCs, but not when FCCs cannot take advantage of a piece of itself that it sends away. A solution would be that in the other situations, when there is no proleptic DP or demonstrative, the clause still is active by means of a vicarious element, which, I claim, is a null expletive *pro*. This resembles Stroik's (1996) analysis, in which FCCs with expletives are deemed to be *in situ* and the expletives are motivated by Case-requirements. There are differences though. First, for the Classical Greek facts, I supposed long-distance Agree, so there is no locality requirement between v° and its goal, contrary to English. Second, and more importantly, the expletive must be null.

Null expletives have already been supposed. For example, Chomsky (1995) suggested that "the expletive can be *pro*, if the language permits null subjects" (p. 123). Likewise, for Stroik's (1996) analysis to be uniform, a null expletive should also be posited in English cases of seemingly extraposed FCCs with no overt expletive *it*. Gluckman (2021) entertains this idea to explain agreement facts in Logoori (Bantu), although he ultimately rejects it in favor of a direct agreement process between T° and the clause, but both accounts seem valid. Finally, Bjorkman and Zeijlstra (2019: 563-564) reach the same conclusion that a "null expletive *pro*" might be a necessary last resort for establishing a long-distance relation between v° and its object. We saw that they distinguish between two operations: Upward Agree (for feature-checking) and Downward Agreement (for feature-valuation), the latter presupposing the former. The cases of valuation with no visible checking are problematic for them, which is why they suppose that checking still takes place, but *via* a null *pro* in Spec,TP. Note that this proposal of a null *pro* resembles Chomsky's (1981; 1986a) claim that any DP must belong to a chain that contains just one θ -marked element and one Case-marked element, although it minimally differs from it in that *pro* is posited for activation and not for visibility (see Introduction of this section). In fact, for Modern Greek FCCs, Roussou (1991: 97 and fn. 16) also posits a null expletive *pro* when the clause is *in situ* to form a licit chain.

The difficulty with such an analysis is that it is hardly falsifiable. Being null, *pro* has no PF reflex. Being an expletive, it has no LF reflex. Nevertheless, several arguments support it. A null *pro* has overt counterparts in other languages, is posited elsewhere for theory internal reasons, and has further empirical motivations in the language. I now turn to the evidence that bolsters these points.

First, overt expletives are common crosslinguistically in relation with FCCs. Apart for English *it*-that structures (on which see Stroik 1996, among many others), it is found for example in as Julia (43), Hungarian (44) or German (45).³⁷

³⁷ Note that *es* in this German example is distinct from other instances of correlative *es* in which it is not just a placeholder but an anaphoric element, e.g., with *sagen* 'say' (Axel-Tober, Holler, and Krause 2016).

- (43) Julia (Kiemtoré 2023: ex. 2b)
 Adama be **(a)* lɔn [PrP *a* [Pr' ko Awa ye bon lɔ]].
 Adama HAB 3SG know COMP Awa PFV house build
 'Adama knows that Awa has built a house.'
- (44) Hungarian (den Dikken 2017: 2b)
 János *azt* hiszi, **hogy Mari terhes**.
 János it:ACC believes that Mari pregnant
 'János believes that Mari is pregnant.'
- (45) German (Axel-Tober, Holler, and Krause 2016: 49)
 ... weil sie es bedauert, **dass sie kein Abitur hat**.
 because she it regrets that she no A-levels has
 '... because she regrets not having taken A-levels.'

In Julia, Kiemtoré (2023) claims that the expletive *a* is generated in a predication phrase with the clause and then moved to Spec,VP for Case reasons. Den Dikken (2017) also supposes a predication structure for (44), but the verb is its head and the expletive is in its specifier, playing the role of the predicate. É. Kiss (2002: 234-235) reports other analyses in which *azt* first occupies the specifier of the embedded CP or heading a DP in which the CP is enclosed. *Azt* then moves to Spec,VP and plays the role of a placeholder for the clause, which is unable to move. Likewise, in German, the *es* element is analyzed as a correlative expletive or as a determiner forming a constituent with the clause before derivation (see discussion in Frey, Meinunger, and Schwabe 2016).

The second motivation for null expletives is that null pronouns are present elsewhere in the language, including in object positions. In particular, Classical Greek is both subject and object *pro-drop* (Luraghi 2003), although overt pronouns are also possible. Both possibilities are exemplified (in italics) in (46). It is important to note that this phenomenon targets contingent, specific referents and not just general, arbitrary referents, as is usual in languages, as observed in Rizzi (1986).

(46) Lys. 3.38 (Luraghi 2003: ex. 20)

Ei	polloús	échōn	tōn epitēdeíōn	egō,	apantésas	Símōni,
if	a.lot	having	of.associates	I	having.met	Simon:DAT
emachómēn	<i>autô</i>	kai	étypton	<i>autòn</i>		
I.fought	him:DAT	and	I.beated	him:ACC		
kai	edíōkon	<i>pro</i>	kai	katalabōn	<i>pro</i> ágein	<i>pro</i> bía ezétoun,...
and	I.chased	and	having.caught	to.drag	by.force	I.tried

‘If I had taken a lot of my associates, met Simon, fought with him, beaten him, chased him, and tried to catch and drag him by force.’

(46) also buttresses the claim that null *pros* bear Case. There is more evidence for that, for example the agreements that we observe in (47) and (48) between the adjective and the *pro* that the θ -criterion independently requires. In both examples, the adjective *filánthrōpos* ‘kindly’ cannot receive its case (number and gender) if we do not suppose that a null *pro* in the nominative, respectively accusative, is present, with which it agrees.³⁸

(47) Isoc. 2.15

pro.ACC	filánthrōpon	eînai	deî.
	kindly:ACC	be:INF	is.necessary

‘It is necessary to be kindly.’

(48) Arist. *Const. Ath.* 16.2

pro.NOM	filánthrōpos	ên.
	kindly:NOM	was

‘He was kindly.’

Finally, in Classical Greek, there is another indication for a null *pro*, aside from the fact that it belongs to the paradigm of FCC-proxies along with proleptic DPs, demonstratives, and extracted *whP*. When the structure is inserted in a correlative clause, the language must insert a *wh*-expletive, which is, I claim, the *wh*-counterpart of the null *pro*. Crucially, this *wh*-element also agrees in number with the FCC. Consider (49) and (50).

³⁸ See for a full demonstration Creider and Hudson (2006). In their framework (Word Grammar) this underlying subject is viewed as a virtual, unreal subject. Although they convincingly show that it is distinct from PRO (section 6), in my view, they do not that it is from *pro*.

(49) Arist. *EE* 1222b37

Dêlon *hò* epikheiroûmen [~~hè~~ **hóti anagkaîon**]

clear what:ACC.N.SG we.argue that necessary

ek tôn analutikôn (*hóti anagkaîon*).

from the analytics

‘From *the Analytics*, what I’m claiming is clear: This is necessary.’

(50) X. *HG* 2.3.45

[*Há* eîpen [~~hè~~ **hōs egō eimi hoîosaeí pote metabállesthai**]]_i,

what:ACC.N.PL he.said that I am such always one.day to.change.sides

katanoēsate kaì *taûta*_i.

consider:IMP.2PL also this:N.PL

lit. ‘what he said that I was the sort of man that always changes sides, consider it as well.’

‘He said that I was the sort of man that always changes sides. Consider it as well.’

The first construal that comes to mind is that the clause is appended to or predicated of the relative. (49) would mean ‘it is clear that this is necessary, which is what I’m arguing’ or ‘what I’m arguing is clear, namely *that this is necessary*’. This construal is however impossible, since the *hóti*-clause would be a constituent of the matrix and thus obey the clause-finality constraint.³⁹ Instead, the *hóti*-clause must be construed as a constituent of the relative clause. The verb *epikheirēō* ‘argue’ selects for *hóti*-clauses in Aristotle⁴⁰ and *hóti anagkaîon* is located clause-finally within the relative, as expected. Head-internal relative/correlative clauses are frequent in Classical Greek and relative *wh*-determiners usually strand the noun when moving up (51) (cf. Faure 2021b: 19-20). That the *wh*-term and the *hóti*-clause initially form a constituent is inferable from examples like (52).

(51) X. *HG* 1.6.3

pros *haîs* parà Lusándrou élabe ~~haîs~~ **nausí**.

to which:DAT.F.PL from L.:GEN he.received ships:DAT.PL

lit. ‘(in addition) to which ships he received from Lysander.’

‘In addition to the ships that he received from Lysander.’

³⁹ Recall from Section 4.2 that there are alternative positions in the upper field of the matrix, but they are not either where the *hóti*-clause stands in (49) and (50).

⁴⁰ Liddell and Scott (1996: s.v. III).

(52) D. 58.63

Hò *kai* *thaumásion* *estin*
what:NOM.N.SG even surprising:NOM.N.SG is
[~~hò~~ *hóti* *zôntes* *ek* *toû* *sukofanteîn*
 that living from the informing
***oú* *fasi* *lambánein* *apò* *tês* *póleōs*].**
NEG they.say take:INF from the city

‘It is even surprising that while living on informing they say that they don’t receive anything from the city.’

Smyth (1956: 561, §2494) analyzes this kind of sentence as an independent clause (bolded) commented by a relative clause beginning with *hó*. This construal is not possible, however. First, there is no reason why what Smyth considers an independent clause would be introduced by the complementizer *hóti*, which everywhere else introduces a subordinate. Second, the sentence is not linked by any word to the previous sentence, which almost never happens in Classical Greek. If conversely *hó* is analyzed as a *relative connection* device, everything falls into place. To connect two sentences, the language has the capacity of using a relative pronoun at the beginning of a sentence instead of a connector + a demonstrative (Smyth 1956: 560, §2490). So, *hó* here is anaphoric and not cataphoric. The matrix clause is the one whose predicate is *thaumásion*. Moreover, the content of the embedded *hóti*-clause is not new but summarizes the previous discussion and is also anaphoric. Thus, it is more natural to construe *hó* and the *hóti*-clause as forming a whole, from which a *wh*-expletive is extracted. Therefore, this idiosyncratic type of relative clause would be an additional argument in favor of an expletive being present in Classical Greek FCCs and making them active. It receives support from crosslinguistic evidence of *wh*-expletives in German, Hindi and Hungarian, which is liable to a similar analysis (Horvath 1997). The reason why the phenomenon is limited to *wh*-questions in these languages and to relative clauses in Classical Greek is left for future research.

5.5 Interim conclusion

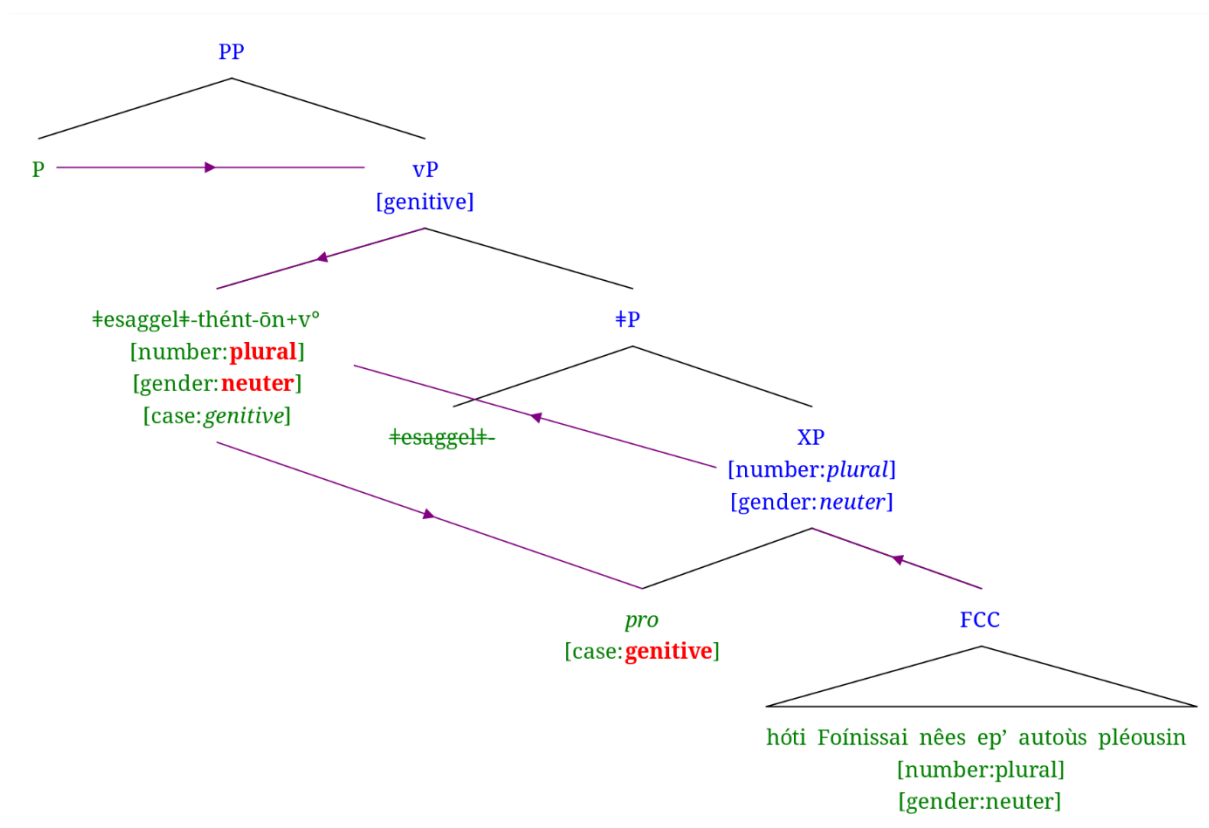
In the present section, we have seen that, although Classical Greek FCCs are not active elements, they are able to participate in the derivation because they have active proxies that bridge the gap between functional heads in search for ϕ -features and the target that they constitute. This analysis comes at the cost of positing a null expletive when there is no overt proxy, but we saw that there are good crosslinguistic and language-internal arguments in favor of this stance. Finally, this allows us to maintain a conservative view of agreement, which is long-distance and downward-playing and where Case-valuation is parasitic onto ϕ -feature valuation (Chomsky 2000; 2001). The division of labor is as such, in the spirit of the chains (Chomsky 1986b, a.o.): The FCC carries the ϕ -features and takes on the θ -role, the proxy activates the structure, meaning that when the proxy is a null expletive, it does not bear ϕ -features.

Before I close this section, I want to give a full view of what a derivation with a covert expletive looks like, going back to (39), repeated here, an example where agreement between the *hóti*-clause and *v*^o is overt. Consider (53) and the derivation that is described below it.

(39) Th. 1.116.3

esaggelthéntōn **hóti**Foínissai nêes ep' autoùs pléousin.
 announce:PTCP.AOR.PASS.N.PL.GEN that Ph. ships:NOM towards them navigate:3PL
 lit. 'having been announced that Phoenician ships were navigating towards them.'

(53) Analysis of (39)



(the arrows indicate feature-percolation or valuation under agreement)

1. A null expletive *pro* is generated with the FCC and form a constituent XP. I remain agnostic as to the nature of XP and their relation (but see the discussion around examples (43)-(45) for hypotheses). *pro* has a case-feature and is thus active for agreement;
2. The ϕ -features [plural] and [neuter] of the FCC percolate to XP;
3. I consider that the verbal form starts its life as a root (noted between two ‡), as in Distributed Morphology (Halle and Marantz 1993), here *esaggel* 'announce';
4. The root *esaggel* selects and merges with XP;
5. *v*^o selects and is merged with ‡P;

6. The root *esaggel* moves up to v° and gets categorized by it as verbal;⁴¹
7. v° enters in relation with *pro* in Spec,XP, which makes XP active for v° ;
8. v° agrees with XP;
9. XP values v° 's ϕ -features;
10. P selects and is merged with vP;
11. The genitive case of the absolute genitive and the participle *esaggelthéntōn* is attributed by a null P to its sister vP and from there the whole structure is imbued with it.⁴²

6 Conclusion

In this article, we engaged in a discussion around Case-marking of FCCs. As complements to a verb, they should not receive a different treatment from other arguments. Argumental DPs are usually taken to be active and enter in relation with functional heads thanks to their unvalued Case-feature. In absence thereof, they would be inaccessible and v° and T° wouldn't agree with them and v° 's and T° 's ϕ -features would remain unvalued.

We tackled this issue from the angle of a particular language, Classical Greek. We showed that

- (1) Its FCCs do not carry Case (no CaseP is projected) and are not active, so should not be able to agree with functional heads;

Notwithstanding,

- (2) They are DPs, endowed with ϕ -features and that they do agree with v° and T° , and value their ϕ -features;

We then focused on the theoretical issue that this paradox represents and uncovered that Classical Greek FCCs, albeit not active themselves, use active proxies to enter into a relation with functional heads and value their ϕ -features. This comes at the cost of positing null expletives in some cases, for which abundant indirect evidence was nevertheless provided. This proof represents one of the most important contributions of the article.

More support could come from interpretation. Although it is usually assumed that expletives are devoid of LF effects, recent work pleads for the contrary (Tsiakmakis and Espinal 2022). As far as Classical Greek is concerned, the structures in which null expletives are supposed are in contrast to those in which the FCC is (post-syntactically) moved to a discourse position and those in which the proxy is overt. When the proxy is the demonstrative *toûto*, Narrow Focus bears on the FCC (Bertrand 2010: 308-3010), when a proleptic DP is extracted out of it, the FCC is informationally demoted (Panhuis 1984). Null expletives would be the indication that the FCC is a subpart of a focal domain. This speculation is left for future research.

Several other questions will have to be investigated in the future as well. The main one is the crosslinguistic validity of the view that was defended here. We observed that in the languages in which FCCs do actively participate in the derivation, they are equipped with additional material, such as Case-marking or heavy complementizer meaning 'saying'. These facts are confirmation for my approach. In contrast, in many languages, such as English, FCCs tend to be clause-final and not participate in the derivation. As hinted at in Section 5.4, we expect the

⁴¹ *Esaggelthéntōn* is decomposed into *esaggel-thé-nt-ōn*: root-pass-ptcp-genitive; In the tree, I ignore the passive and participle morphemes and projections, which enter the derivation above vP.

⁴² See the discussion around ex. (1).

proxy approach to extend to them, even though they do not exhibit as many proxies as Classical Greek.

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